

Module Code:	CS5003NI
Module Title:	Data Structures and Specialist Programming
Module Leader:	Mr. Sudip Dahal (Islington College)

Coursework Type:	Individual
Coursework Weight:	This coursework accounts for 30% of your total module grades.
Submission Date:	
Coursework is given out:	Week 20
Submission Instructions:	Submit the following to the Islington College's MST portal before 01:00 PM on the due date: • A report (document) in .pdf format in the MST portal or through any medium which the module leader specifies.
Warning:	London Metropolitan University and Islington College takes Plagiarism seriously. Offenders will be dealt with sternly.

PLAGIARISM

You are reminded that there exist regulations concerning plagiarism. Extracts from these regulations are printed overleaf. Please sign below to say that you have read and understand these extracts:

Extracts from University Regulations on Cheating, Plagiarism and Collusion

Section 2.3: "The following broad types of offence can be identified and are provided as indicative examples

Cheating: including taking unauthorised material into an examination; consulting unauthorised material outside the examination hall during the examination; obtaining an unseen examination paper in advance of the examination; copying from another examinee; using an unauthorised calculator during the examination or storing unauthorised material in the memory of a programmable calculator which is taken into the examination; copying coursework.

Falsifying data in experimental results.

Personation, where a substitute takes an examination or test on behalf of the candidate. Both candidate and substitute may be guilty of an offence under these Regulations.

Bribery or attempted bribery of a person thought to have some influence on the candidate's assessment.

Collusion to present joint work as the work solely of one individual.

Plagiarism, where the work or ideas of another are presented as the candidate's own.

Other conduct calculated to secure an advantage on assessment.

Assisting in any of the above.

Some notes on what this means for students:

Copying another student's work is an offence, whether from a copy on paper or from a computer file, and in whatever form the intellectual property being copied takes, including text, mathematical notation, and computer programs.

Taking extracts from published sources *without attribution* is an offence. To quote ideas, sometimes using extracts, is generally to be encouraged. Quoting ideas is achieved by stating an author's argument and attributing it, perhaps by quoting, immediately in the text, his or her name and year of publication, e.g. " $e = mc^2$ (Einstein 1905)". A *reference* section at the end of your work should then list all such references in alphabetical order of authors' surnames. (There are variations on this referencing system which your tutors may prefer you to use.) If you wish to quote a paragraph or so from published work then indent the quotation on both left and right margins, using an italic font where practicable, and introduce the quotation with an attribution.

CONTRACT CHEATING

Contract cheating (also known as assessment outsourcing, commissioning or ghost writing) is when someone seeks out another party, or AI generator service, to produce work or buy an essay or assignment, either already written or specifically written for them or the assignment to submit as their own piece of work.

Contract cheating undermines the integrity of the academic process and devalues the qualifications awarded by the university. Students are reminded that academic integrity is a fundamental principle of our institution. Engaging in contract cheating not only impacts the individual's academic record but also the reputation of the university.

Students are encouraged to seek support if they are struggling with their coursework. The university offers a range of resources, including academic counseling, tutoring services, and workshops on study skills and time management. Utilizing these resources can help students achieve their academic goals without resorting to dishonest practices.

Penalty:

- Failure in the Module: The student must re-register for the same module, and the re-registered module will be capped at a bare pass.
- Ineligibility to Continue on the Course: Where re-registration of the same module, or a suitable alternative, is not permissible, the student will not be able to continue on the course. Additionally, the following penalty will be applied to the student's final award:
 - Undergraduate Honors: The student's final classification will be reduced by one level.
 - Unclassified Bachelors: Downgraded to Diploma in Higher Education.
 - Foundation Degree: Distinction downgraded to Merit; Merit downgraded to Pass; Pass downgraded to Certificate in Higher Education.
 - Masters: Distinction downgraded to Merit; Merit downgraded to Pass; Pass downgraded to Postgraduate Diploma.

Reporting and Consequences:

Instances of contract cheating will be thoroughly investigated, and students found guilty will face the penalties outlined above. It is the responsibility of every student to ensure that their work is their own and to avoid situations that could lead to accusations of academic misconduct.

By adhering to these standards, students contribute to a fair and equitable academic environment, ensuring the value and recognition of their qualifications are maintained.

Deliverables

You are required to submit two components before the submission deadline.

1. Dynamic System:
 - The **project**, including the generated **SQL schema**, must be uploaded to a **GitHub repository**.
2. Report in PDF format.
 - Provided report structure and guidelines in **Task B** should be included.
 - Additionally, in your cover page include **a link to the GitHub repository** in your report, ensuring that the repository is publicly accessible.

Task A: System Development

Please note that the scenario described below cannot be used as your coursework project. You are required to propose a new and original project idea that is distinct from any other projects from students and does not align with the provided project scenario.

Project Scenario

For example, if we propose a **Library Management System (LMS) named "BookTrack"**, as our coursework project, the system would serve as a comprehensive web-based platform to **manage Book-related activities** within an educational institution. This project would allow students to explore key aspects of web development, including **user authentication, role-based authorization, data management, CRUD operations and user interface design**.

The detailed information on system components is explained below:

1. Database Design and Development [5 marks]

- a. Required database design and development with appropriate (real-time) data should be included.
- b. Tables should be create accordingly for users and book information.

2. Admin Dashboard: [10 marks]

- a. The admin dashboard would enable administrators to manage student records by **creating, updating, deleting, and viewing** detailed information such as book data (e.g., title, author, publication year, available status, ISBN) and member data (e.g., name, membership ID, contact details, borrowed books, due dates).

3. Member Portal: [10 marks]

- a. The member portal would provide library members with a platform to **manage their profiles**, allowing them to log in, view, and update their personal information, including contact details and passwords.
- b. The system will include a search feature that allows members to find specific information, such as books, authors, or genres, using a search bar. This functionality will enable members to quickly access relevant details by entering keywords or criteria related to their search.
- c. Data on the member home page, such as available books, borrowed books, and due dates, should be displayed in real time.

4. Authentication & Authorization [5 marks]

- a. The system would employ **secure login protocols** to ensure that only authorized users can access the platform.
- b. Access control should be **role-based**, distinguishing between **administrative users and normal users** i.e., students, enforcing restrictions on functionalities based on the user's role, and proper encryption program.
- c. **Session** can be widely used to complete the **authentication and authorization process**.
- d. **Users** can **reset their password** if they forget it.
- e. If a user enters the **wrong password** too many times, the account will be **temporarily locked**.
- f. **Redirect Management (Filter)**
 - i. The system would efficiently **manage URLs**, directing users to the appropriate pages based on their actions and roles. Unauthorized access attempts would result in redirects to the login page or an error page, while session management would ensure that users remain logged in across different pages until they choose to log out or their session expires.

5. MVC Architecture [5 marks]

- a. The system will be developed using the **Model-View-Controller (MVC)** architecture, which separates the application into three parts.
 - i. The **Model** component manages the data of books, members, librarians, and borrowing record .
 - ii. The **view** component **built with JSP and CSS**, shows the data to users based on their role (e.g., admin, librarian, or member).
 - iii. The **Controller** will process user requests, coordinate with the Model and Service to retrieve or update data, and determine the appropriate View to display.
- b. In addition to the MVC components, the system will utilize packages such as **util and service**, along with other necessary packages, to enhance functionality and

maintainability.

- i. The util package will include utility classes and methods for handling common tasks like validation, date formatting, and other reusable functions.
- ii. The service package will encapsulate business logic, facilitating interactions between the Controller and the data access layers.

6. Frontend Design and Development [5 marks]

- a. The user interface would be built using **JavaServer Pages (JSP)** for dynamic content and **CSS** for styling, ensuring a **responsive** and user-friendly design that accommodates a range of devices, from desktops to mobile phones. Use of frameworks like bootstrap and others are prohibited. **You can use CSS media query and flexbox properties for responsive design.**
- b. JavaScript should be only used as needed for **enhancing the design and user experience** of the application. The core functionality and server-side logic will be managed by the Java-based components of the system.

7. Additional Features: [5 marks]

The system can improve with the inclusion of:

- a. **About page** to provide information about the institution and the purpose of the LMS.
- b. **Contact page** offering support details and a form for inquiries and so on
- c. **Most recently added books.**
- d. **Fine management.**

Note:

While this scenario serves as a detailed example of a web application project, students are not permitted to use it for their coursework. Instead, they must propose and develop a unique project centered around an ethical topic, ensuring that each project is distinct. These projects should incorporate similar technical components, such as MVC architecture, role-based access control, and responsive design, providing students with practical experience in web development using

Java, JSP, CSS, J2EE and MySQL. The coursework aims to enhance students' skills in system design, implementation, and problem-solving, while also considering ethical implications.

Other Program Requirements: [5 marks]

1. Validation and Exception Handling:

- During the data registration process, it is imperative to implement validation checks to verify the existence of an account. This can be done by cross-referencing a unique identifier, such as a phone number, to ensure that the account does not already exist.
- Proper exception handling mechanisms must be in place to manage potential errors gracefully in entire system.

2. Programming Styles:

- The Java code should follow standard naming conventions and include meaningful comments to enhance code readability and maintainability. These practices are crucial for maintaining clarity in the code and ensuring that it adheres to professional standards.

3. Proper Use of Classes and Methods:

- Object-oriented programming principles should be applied efficiently to reduce code redundancy. This involves utilizing methods and classes effectively to create a clean and modular codebase, emphasizing reusability and maintainability.

4. Use of Proper Messages:

- When creating a new account, the system must provide appropriate prompts if any entered information is invalid. For example, if a user enters numerical data in the full name field, the system should alert the user with a clear and specific error message, guiding them to correct the input.

Task B: Report

Word Count: 15,000 (word count starts from introduction till conclusion.)

Note that you must appropriately follow the report structure and styles. [6 marks]

1. Introduction

This section should include a clear and concise introduction to the project, covering the following elements:

- **Title:** Provide a descriptive title for your project.
- **Purpose:** Explain the purpose of the project and why it is being developed.
- **Audience:** Identify the intended audience for your application.
- **Aims and Objectives:** Outline the main aims and objectives that the project seeks to achieve.

2. Problem Statement [4 marks]

This section should outline the limitations/problems of existing systems and explain how your proposed method addresses these issues.

3. Wireframe [8 marks]

- Present the wireframes of your application alongside your prototype designs, showing the layout and structure of the user interface. This should include all major screens, such as the login page, dashboard, and any other key pages like Search, Reports.

4. Java Classes and Methods Explanation [6 marks]

- Provide a concise explanation of the Java classes used in your application. This section should include:
 - i. **Class Diagram:** A visual representation of the classes, their relationships, and interactions.
 - ii. **Methods Explanation:** For each Java class, list the methods used and explain their purpose and functionality.

5. Test Cases [6 marks]

- Create a test case table with following columns:
 - Test Case ID

- Test Description
- Input Data
- Expected Output
- Actual Output
- Cover Functional testing, Validation testing, Exception handling testing and UI testing

6. Coursework Development [6 marks]

Document the development process of your coursework, including the tools used to complete the system. For each tool, provide:

- **Tool Description:** What the tool is and its primary purpose.
- **Usage:** How the tool was used in the development process.
- **Evidence:** Include screenshots or other evidence demonstrating the use of the tool.
- You can take **Database Tables and ER Diagrams** as an example to start with.

- **List of Database Tables:**

Identify and list the tables utilized within your application, such as

- **Book** - stores information about each book (title, author, ISBN, availability, category)
- **Members** – stores member details (name, contact, membership ID).
- **BorrowRecords** – tracks which member borrowed which book, issue date, and due date.
- **Reservations** – stores book reservations by members.
- **Fines** – records fines for overdue books.

- **Relational Schemas(Database Schema):**

Provide a comprehensive Relational schema diagram that illustrates the relationships between these tables, highlighting primary and foreign keys.

- **Tool Utilized:**

Specify the tool employed, such as XAMPP, and its primary function in designing and managing the database schema.

- **Application of the Tool:**

Detail how the tool was used to develop a complete database and generate the ER diagram, ensuring the integrity and efficiency of the data structure.

- **Evidence:**

Include relevant evidence, such as screenshots of the database tables and the ER diagram, to substantiate your implementation process.

7. Critical Analysis [4 marks]

Conduct a critical analysis of the system, evaluating its performance and identifying any challenges or difficulties encountered during development. This section should include:

- **System Breakdown:** Discuss any issues, errors, or obstacles faced during development, supported by evidence such as screenshots.
- **Evaluation:** SWOT(Strength ,Weakness, Opportunities, Thread) analysis.
- **Comparison with other existing systems:** Identify problems of existing system and how did you solve these problem
- **Performance Metrics:** Analyze how fast searches are, and how quickly data loads.

8. Conclusion and Future recommendation [4 marks]

Summarize the key findings from your coursework. Reflect on what you have learned, any challenges faced, and how you overcame them. You should discuss the limitation, potential improvements and additional features that could enhance your proposed system beyond the current implementation.

9. References (Bibliography) [4 marks]

List all the sources referenced throughout your report in a properly formatted bibliography.

10. Appendices [2 marks]

Additional diagrams, code screenshots.

Important Details:**1. Mandatory Use of Technologies:**

Failure to incorporate required technologies such as Java, Java EE, MySQL, or JSP may result in an automatic failure of the coursework.

2. Program Functionality:

Ensure that your program operates correctly on at least two different PCs prior to submission to avoid technical issues.

3. Viva Attendance:

Non-attendance at the viva examination will lead to a failure in the system evaluation.